Journal of Abnormal and Social Psychology
1961, Vol. 63, No. 2, 381-385

THE STIMULATING VERSUS CATHARTIC EFFECTS OF A VICARIOUS AGGRESSIVE ACTIVITY

SEYMOUR FESHBACH
University of Pennsylvania

THE present study is concerned with the complex effects of participation in a presumably vicarious aggressive activity upon subsequent aggressive behavior. A number of studies have demonstrated that the expression of aggression—whether directly or in symbolic form—results in a lowering of subsequent aggression (Berkowitz, 1960; Feshbach, 1955; Pepitone & Reichling, 1955; Rosenbaum & de Charms, 1960; Thibaut & Coules, 1952). However, there is also experimental evidence to the effect that aggressive activity has a stimulating effect upon the manifestation of other aggressive acts (Feshbach, 1956; Kenny, 1953); that is, aggression may breed aggression.

Since both possibilities—reduction and stimulation—have been experimentally observed, the pertinent issue then is under what conditions a vicarious aggressive act increases and under what conditions it decreases the probability of subsequent aggressive behavior. One such condition suggested by differences in procedure between the studies that obtained evidence of a cathartic effect and those demonstrating a stimulating effect is the emotional state of the subject at the time the aggressive act is performed; that is, if the subject is angry at the time he engages in the aggressive activity, he can then use the act to satisfy and thereby reduce his hostility.

The general hypothesis is suggested that in order for an activity to have drive reducing properties, components of the drive must be present or evoked during performance of the activity; that is, there must be some functional connection between the vicarious act and the original drive instigating conditions. While it is undoubtedly true that the vicissitudes of life will arouse hostilities that cannot be directly discharged, it does not follow that any indirect aggressive act will have the property of reducing hostility that has been evoked under markedly different circumstances. According to the present view, a child's anger toward its mother will not be reduced by an aggressive act toward a doll figure unless its anger toward the mother is aroused when the aggressive act is performed. The evocation of anger may not be a sufficient condition—the doll figure may have to be similar to the mother—but it is probably a necessary condition for drive reduction to take place.

If the subject is not hostile at the time of participating in an aggressive act, his subsequent aggressive behavior will not merely remain unaffected but is very likely to be increased. An increase in aggression following a vicarious aggressive act could result from a number of different processes: a reduction in inhibition or aggression anxiety, reinforcement of aggressive responses, and finally conditioned stimulation of aggressive drive and/or aggressive responses.

On the basis of the foregoing considerations, the following hypotheses are proposed: Participation in a vicarious aggressive act results in a reduction in subsequent aggressive behavior if aggressive drive has been aroused at the time of such participation; if aggressive drive has not been aroused at the time of participation in a vicarious aggressive act, such participation results in an increase in subsequent aggressive behavior.

METHOD

The experimental procedure consisted of arousing a subject's aggressive drive before participation in a vicarious aggressive act or before participation in a neutral act and then obtaining measures of aggression subsequent to these interpolated activities. The variation in level of aggression was accomplished by means of an insult versus noninsult condition and the variation in the interpolated activity consisted of exposure to a fight film versus a neutral film.

Subjects

The subjects were male college student volunteers who were assigned at random to one of the four treatment groups generated by the two experimental variables. One hundred and one subjects were used in the study, with approximately equal numbers in each experimental condition. The subjects were seen in

small groups by the experimenter so that nine experimental sessions in all were held, three for the Noninsult Fight Film condition and two sessions for each of the other experimental conditions.

Procedure

Insult versus Noninsult condition. Subjects assigned to the insult groups were subjected to a number of unwarranted and extremely critical remarks. These comments essentially disparaged the intellectual motivation and the emotional maturity of the students.[1] Previous studies (Feshbach, 1955; Gellerman, 1956) have provided abundant evidence that this technique successfully arouses hostility toward an experimenter. Subjects assigned to the Noninsult condition were given standard test instructions.

Aggressive Film versus Neutral Film condition. Subjects in the Insult and Noninsult groups then witnessed either a Fight Film or a Neutral Film. The Fight Film consisted of a film clip of a rather exciting prize fight sequence taken from the motion picture *Body and Soul* while the neutral film depicted the consequence of the spread of rumors in a factory. The duration of each of the films was approximately 10 minutes.

As a rationale for the presentation of the film, the subjects were told before the film was presented that they would be asked to judge the personality of the main character in the film. Following the completion of the film, each subject indicated his impression of the personality of the hero of the film on a questionnaire provided for that purpose.

Dependent Measures of Aggression. All subjects were given a modified word association test which, in a previous study (Gellerman, 1956) had been shown to be sensitive to differences in the arousal of aggression. The test involves the presentation of five aggressive words interspersed among six neutral stimuli as follows: wash, choke, travel, massacre, walk, murder, relax, stab, sleep, torture, listen. The subjects are asked to give in written form a series of associations to each word. The stimuli are presented orally and also visually, the experimenter holding up a 5″ × 8″ card on which the stimulus word is printed. The subject's Aggression score is based on the number of aggressive word associations among the first 10 responses to each of the aggressive stimulus words. The maximum score that can therefore be obtained on this measure is 50.

Subsequent to the administration of the word association test, the first experimenter left the room having presumably completed the study. A second experimenter then entered and informed the subjects that the psychology department wished to assess students' opinions of the value of participating in psychological experiments. A questionnaire was then administered dealing with the subjects' attitudes toward the experimenter and with their evaluation of the experiment. The questionnaire which consists of six items, each of which has six alternatives, is described in more detail in a previous study (Feshbach, 1955). It is scored so that the least aggressive choice for a particular item is given a score of 1 and the most aggressive choice, a score of 6.

[1] The author wishes to express his gratitude to Abraham Wolf for his competence and courage in carrying out this phase of the experiment.

RESULTS

By hypothesis, it was predicted that the Insult group exposed to the Fight Film would manifest *less* subsequent aggression on each of the two measures of aggression than the Insult group exposed to the Neutral Film while the Noninsult group exposed to the Fight Film would display *more* subsequent aggression than the Noninsult group exposed to the Neutral Film. The word association data bearing upon these predictions are presented in Table 1. The mean differences are in accordance with expectation, the Insult-Fight (IF) Film group responding with fewer aggressive associations than the Insult-Neutral (IN) Film group and the Noninsult-Fight (NIF) Film group responding with more aggressive associations than the Noninsult-Neutral (NIN) Film group. The results of an analysis of variance of the data indicate that the interaction between the Insult and the Film variable is statistically significant. The difference between the IF Film and the IN Film groups falls short of the 5% confidence level, the value of t being 1.9. The difference between the NIF Film and NIN Film groups

TABLE 1

A. MEAN AGGRESSIVE WORD ASSOCIATION RESPONSES OBTAINED UNDER EACH EXPERIMENTAL CONDITION

Drive (D)	Film (F)	
	Fight (N)	Neutral (N)
Insult	24.5 (25)[a]	28.9 (21)
Noninsult	27.7 (29)	25.3 (25)

B. SUMMARY OF ANALYSIS OF VARIANCE OF AGGRESSIVE WORD ASSOCIATION RESPONSES

Source	SS	df	MS	F
D	8.93	1	8.93	
F	38.43	1	38.43	
DF	291.59	1	291.59	4.58*
Within	6,111.80	96	63.66	
Total	6,450.75	99		

[a] The word associations of one subject were not scored due to illegibility.

* $p < .05$.

TABLE 2
DISTRIBUTION OF AGGRESSIVE WORD ASSOCIATION RESPONSES FALLING ABOVE AND BELOW THE MEDIAN AS A FUNCTION OF INSULT FIGHT FILM AND INSULT NEUTRAL FILM TREATMENTS

Treatment	<27	>27
Insult Fight Film	17	8
Insult Neutral Film	10	19

Note.—$\chi^2 = 6.02$; $p < .05$.

TABLE 3
A COMPARISON OF MEAN SCORES ON THE AGGRESSION QUESTIONNAIRE

	Insult-Fight (IF) ($N = 26$)	Insult-Neutral (IN) ($N = 29$)	Noninsult-Fight (NIF) ($N = 20$[a])	Noninsult-Neutral (NIN) ($N = 25$)
M	14.6	19.5	13.7	15.0
σ	3.72	3.90	2.52	2.95

Note.—IF–IN = 4.7; $p < .01$.
[a] One subject failed to complete questionnaire.

yields a t value of approximately 1 which is clearly not significant.

The contrast between the IF Film and IN Film groups is more sharply delineated by a simple median split. The chi square for the fourfold table presented in Table 2 is 6.02 which yields a p value of $<.02$. The word association data, then, indicate that under conditions of anger-arousal, witnessing a fight film results in a lowering of aggression. However, the hypothesized stimulating effect of an aggressive film under nonaroused conditions is not borne out by the data.

The questionnaire data are presented in Table 3. Because of the lack of homogeneity of variance between the IN and NIF Film groups, separate comparisons were made between pertinent groups and, in these comparisons, the variances of the respective distributions are not reliably different. As was the case with the word association data, the IF Film group displays significantly less aggression on the questionnaire than does the IN group. The difference between the Noninsult groups is not in the predicted direction but is small and unreliable.

The difference in subsequent aggressive attitudes between the insulted group exposed to the fight film and the insulted group exposed to the neutral film is further illustrated by a simple median split. The chi square for the fourfold table presented in Table 4 is 15.1, which is significant at less than the .001 level.

TABLE 4
DISTRIBUTION OF AGGRESSION QUESTIONNAIRE SCORES FALLING ABOVE AND BELOW THE MEDIAN AS A FUNCTION OF INSULT FIGHT FILM AND INSULT NEUTRAL FILM TREATMENTS

Treatment	<17.5	>17.5
Insult Fight Film	20	6
Insult Neutral Film	7	22

Note.—$\chi^2 = 15.1$; $p < .001$.

DISCUSSION

The experimental results are consistent with the hypothesis that the drive reducing effect of a vicarious aggressive act is dependent upon the aggressive state of the subject at the time of the vicarious aggressive activity. Witnessing the prize fight film resulted in a significant relative decrement in aggression in comparison to witnessing the neutral film only for those subjects in whom aggression had been previously aroused by the insulting comments of the experimenter. The predicted increase in aggression for the noninsulted subjects following exposure to the fight film did not occur, however. Each of these two outcomes warrants further comment.

With regard to the difference between the two Insult groups in subsequent aggression, a possible alternative to a catharsis or drive reduction hypothesis is one that assumes that guilt or revulsion stimulated by the fight film is the primary mechanism responsible for the lowered aggression. Berkowitz (1958, 1960) has strongly argued for such an explanation of a reduction in aggressive behavior following an aggressive act. However, it must be noted that the evidence for a guilt or inhibition process is most indirect and inferential.

With regard to the present study, the guilt alternative is certainly possible, although, for various reasons to be suggested below, not a likely one. If guilt arousal were a ubiquitous process, occurring whenever people are given the opportunity to indulge in aggressive fantasies, then the fight film should similarly

have inhibited the aggressive response output of the Noninsult group. The possibility still remains that guilt arousal can account for the aggression reducing effects of fantasy under conditions where aggression has recently been stimulated, as in the Insult condition. As a check on whether the lowered aggression of the IF Film group was due to some inhibitory factor, the word associations were scored for defensiveness. A previous study of the effects of inhibition upon aggressive word associations has shown that when inhibition is experimentally aroused, the number of aggressive responses decreases while the number of defensive responses increases (Gellerman, 1956). While, in the present study, a difference was observed in the number of aggressive associations, the difference between the two Insult groups in the number of defensive associations was negligible and insignificant. The absence of an increment in defensive responses, while not decisive since the experiment cited employed an inhibition procedure more closely resembling fear rather than guilt, is more consistent with a drive reduction rather than guilt explanation of the decrease in aggression following the exposure of the insulted subjects to the Fight Film.

The problem remains of accounting for the failure to obtain the expected increase in aggression in the Noninsult group. One possible reason is the limitation of the questionnaire instrument as a measure of aggression. Although one's preference for or attitude toward another person is frequently used as an index of aggression, as was the case in the present experiment, dislike and aggression are not equivalent dimensions. At the extreme, aversion and aggression are likely to be strongly correlated but within moderate ranges of feeling, the association between dislike and aggression may well be negligible. For this reason, the word association measure is probably a better instrument than the attitude questionnaire for detecting changes in aggression in the noninsulted groups. However, although the relative increment in aggressive associations following exposure of the Noninsult group to the Fight Film was in the predicted direction, it was not statistically significant. Whether this failure to obtain evidence of a stimulating effect of a vicarious aggressive activity under relaxed emotional conditions is due to inadequacies in the theoretical analysis or to limitations in the methods utilized cannot be ascertained from the present data.

On the other hand, the data consistently reflect the dependence of the drive reduction effect upon the arousal of aggression at the time the subject is engaging in the vicarious aggressive activity. Presumably vicarious aggressive acts do not willy-nilly serve as outlets for aggressive motivation. This latter process warrants further attention. Aggression is not an ever-present tension system pervading all of an individual's activities. Like other acquired motives, its appearance is very much dependent upon situational factors; and, the more specific the category of objects toward which the aggression is directed, the narrower is both the range of stimuli that can elicit the motivation and the range of situations that can serve as substitute outlets for the aggression.

What would appear to be a relatively simple matter—the effects of a vicarious aggressive activity upon subsequent aggressive behavior—is in actuality a quite complex process. The present study has examined the influence of the drive state of the organism upon this process. Beyond the requirement of replication in a variety of situations, further research is needed to establish the extent to which other variables determine the effects of so-called vicarious aggressive activities and to establish the precise mechanism by which the performance, direct or vicarious, of an aggressive act influences subsequent aggressive behavior.

SUMMARY

Studies of the effects of a presumably vicarious aggressive activity upon subsequent aggressive activity suggest that under certain conditions the activity will tend to increase, and under other conditions decrease, the probability of subsequent aggressive behavior. The purpose of this experiment was to study the effects of one such condition—namely, the emotional state of the subject at the time the vicarious aggressive activity is performed. Specifically, it was proposed that a vicarious aggressive activity results in a *reduction* in subsequent aggressive behavior if the subject is emotionally aroused at the time he is engaging in this activity, but if anger has not

been aroused, the activity results in an *increase* in subsequent aggressive behavior. The two independent variables manipulated in the study consisted of an Insult versus Noninsult condition and an Aggressive Film versus Neutral Film condition. One hundred and one college students were assigned at random to the four treatment groups generated by the two experimental variables. The subjects met the experimenter in small groups so that nine experimental sessions in all were held. Subjects assigned to the Noninsult condition were given standard test instructions while subjects in the Insult groups were subjected to a number of unwarranted and extremely critical remarks. The subjects then witnessed either an Aggressive Film or a Neutral Film. The former consisted of a film clip depicting a prize fight sequence while the latter depicted the consequences of the spread of rumors in a factory. They were then administered a word association test and under the guise of a departmental assessment of the value of students' serving as experimental subjects, a second experimenter administered a questionnaire dealing with the subjects' attitudes toward the first experimenter and with their evaluation of the experiment. The degree of aggression manifested on the attitude questionnaire and the number of aggressive responses on the word association test constituted the dependent measures.

A significant interaction in the predicted direction was obtained for the Word Association measure—the Insult-Aggressive Film group responding with fewer aggressive associations than the IN Film group, and the Noninsult-Aggressive Film group responding with more aggressive associations than the NIN Film group. A similar significant difference between the two Insult groups was found on the attitude questionnaire, but the difference between the two Noninsult groups on this measure was unreliable and was not in the predicted direction.

The results were interpreted as being consistent with a drive reduction theory, although an inhibitory process (guilt arousal) cannot be excluded by the evidence at hand. The dependence of the *aggression* reducing effect of exposure to a film depicting violent activity upon the prior or simultaneous arousal of aggressive drive was stressed.

REFERENCES

BERKOWITZ, L. The expression and reduction of hostility. *Psychol. Bull.*, 1958, **55**, 257–283.

BERKOWITZ, L. Some factors effecting the reduction of overt hostility. *J. abnorm. soc. Psychol.*, 1960, **60**, 14–22.

FESHBACH, S. The drive-reducing function of fantasy behavior. *J. abnorm. soc. Psychol.*, 1955, **50**, 3–11.

FESHBACH, S. The catharsis hypothesis and some consequences of interaction with aggressive and neutral play objects. *J. Pers.*, 1956, **24**, 449–462.

GELLERMAN, S. The effects of experimentally induced aggression and inhibition on word association response sequences. Unpublished doctoral dissertation, University of Pennsylvania, 1956.

KENNY, D. T. An experimental test of the catharsis theory of aggression. Unpublished doctoral dissertation, University of Michigan, 1953.

PEPITONE, A., & REICHLING, G. Group cohesiveness and the expression of hostility. *Hum. Relat.*, 1955, **3**, 327–337.

ROSENBAUM, M. E., & DE CHARMS, R. Direct and vicarious reduction of hostility. *J. abnorm. soc. Psychol.*, 1960, **60**, 105–111.

THIBAUT, J. W., & COULES, J. The role of communication in the reduction of interpersonal hostility. *J. abnorm. soc. Psychol.*, 1952, **47**, 770–777.

(Received October 19, 1960)